

Blazor Component Virtualization

1. Introduction

Blazor Component Virtualization is a technique used to improve performance when displaying large amounts of data. Instead of rendering all items at once, Blazor only renders the items currently visible on the screen. As the user scrolls, Blazor dynamically updates the rendered items.

Virtualization helps reduce:

- Memory usage
- CPU rendering cost
- Slow loading
- Laggy scrolling

2. Virtualize Component

The Virtualize component is a built-in Blazor component used to virtualize lists and UI elements.

Example:

```
<Virtualize Items="items" Context="item">
  <div>
    @item
  </div>
</Virtualize>
```

The component only renders visible items instead of rendering the entire collection.

3. ItemProvider Delegate

ItemProvider is used to load data dynamically when needed instead of loading all data at once.

It is useful for:

- APIs
- Databases
- Large datasets
- Infinite scrolling systems

4. Placeholder

Placeholder displays temporary content while data is loading.

Example:

```
<Placeholder>
  <div>
    Loading...
  </div>
</Placeholder>
```

Placeholder improves user experience and prevents empty blank areas while waiting for data.

5. EmptyContent

EmptyContent displays custom UI when no data exists.

Example:

```
<EmptyContent>
  <div>
    No items found.
  </div>
</EmptyContent>
```

This improves usability and provides clear feedback to users.

6. ItemSize

ItemSize defines the approximate height of each item in pixels.

Example:

```
ItemSize="80"
```

This helps Blazor calculate scrolling positions more accurately and improves virtualization performance.

7. OverscanCount

OverscanCount renders extra items outside the visible area.

Example:

```
OverscanCount="3"
```

This reduces loading flashes during fast scrolling and creates smoother scrolling behavior.

8. State Changes

Virtualized components automatically update when data changes.

Examples:

- Adding items
- Removing items
- Filtering search results

Blazor rerenders the visible items when component state changes.

9. Root-level Virtualization

Root-level virtualization uses the browser page scroll instead of a custom scroll container.

It is useful for:

- Full-page feeds
- Social media timelines
- Large dashboards

10. Keyboard Scroll Support

Keyboard scroll support allows users to navigate virtualized content using the keyboard.

Examples:

- Arrow keys
- Page Up
- Page Down

This improves accessibility and usability.

11. Control the Spacer Element Tag Name

Virtualize uses spacer elements to maintain correct scrollbar height while only rendering visible items.

Spacer elements simulate the height of non-rendered items.

12. Advanced Styles and Scroll Detection

Virtualized components support custom styling and scrolling behavior.

Examples:

- Custom scroll containers
- Custom item styles
- Scroll event handling

Blazor detects scroll position and updates visible items dynamically.

13. Advantages of Virtualization

Advantages include:

- Faster rendering
- Lower memory usage
- Better performance
- Smoother scrolling
- Better user experience

Virtualization is especially useful for applications that display thousands of items.

14. Conclusion

Blazor Component Virtualization is an important performance optimization feature in ASP.NET Core Blazor. It improves application responsiveness by rendering only visible items. Features such as `ItemProvider`, `Placeholder`, `EmptyContent`, and `OverscanCount` help create efficient and scalable applications.